StockSense AI

Sales forecasting and stock planning

Sunrise Mini Market is a fictional retail case

The retail ordering problem

A manager must decide how much to order before next week's sales are known.

Too little stock can mean missed sales.

Too much stock ties up money.

The app gives the manager a consistent estimate to check before ordering.

Objective and scope

Forecast seven days of sales for five products, then show a transparent order suggestion.

One fictional retailer mapped to dataset Store 1

Product 1–5 map to anonymous dataset Item 1–5

Stock and supplier values are entered for the demonstration

Sales dataset

913,000

rows in the source CSV

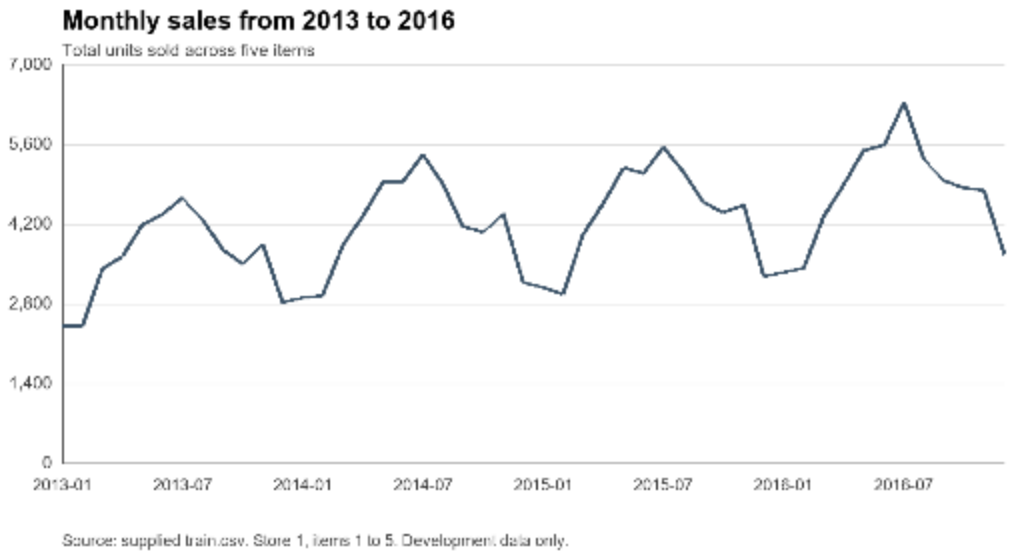
9,130

rows in our selected scope

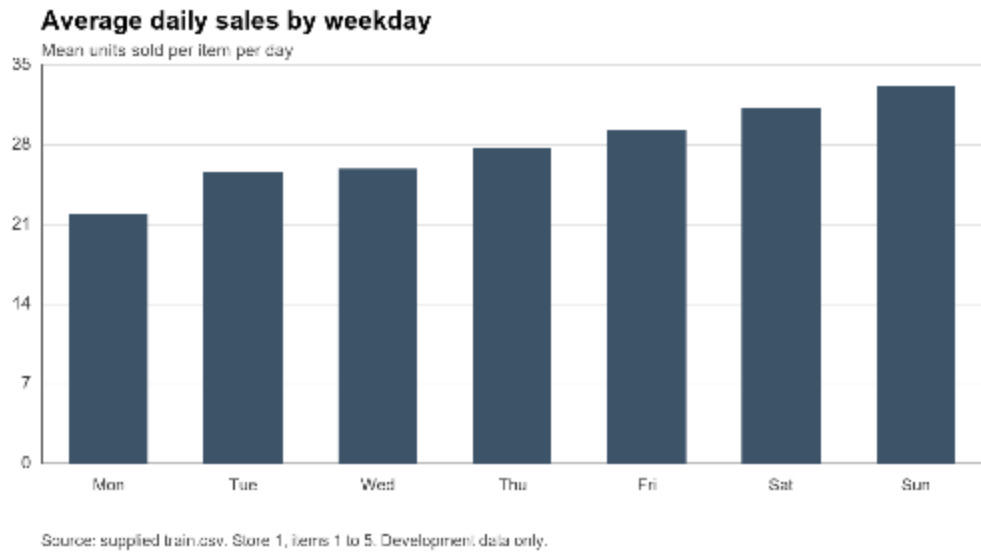
Daily sales, 2013–2017. The file has dates, store IDs, item IDs and units sold.

The dataset has no product names, stock levels, prices or supplier records.

Sales patterns in development data



Monthly totals change over time.



The average differs by weekday.

Forecast methods and time split

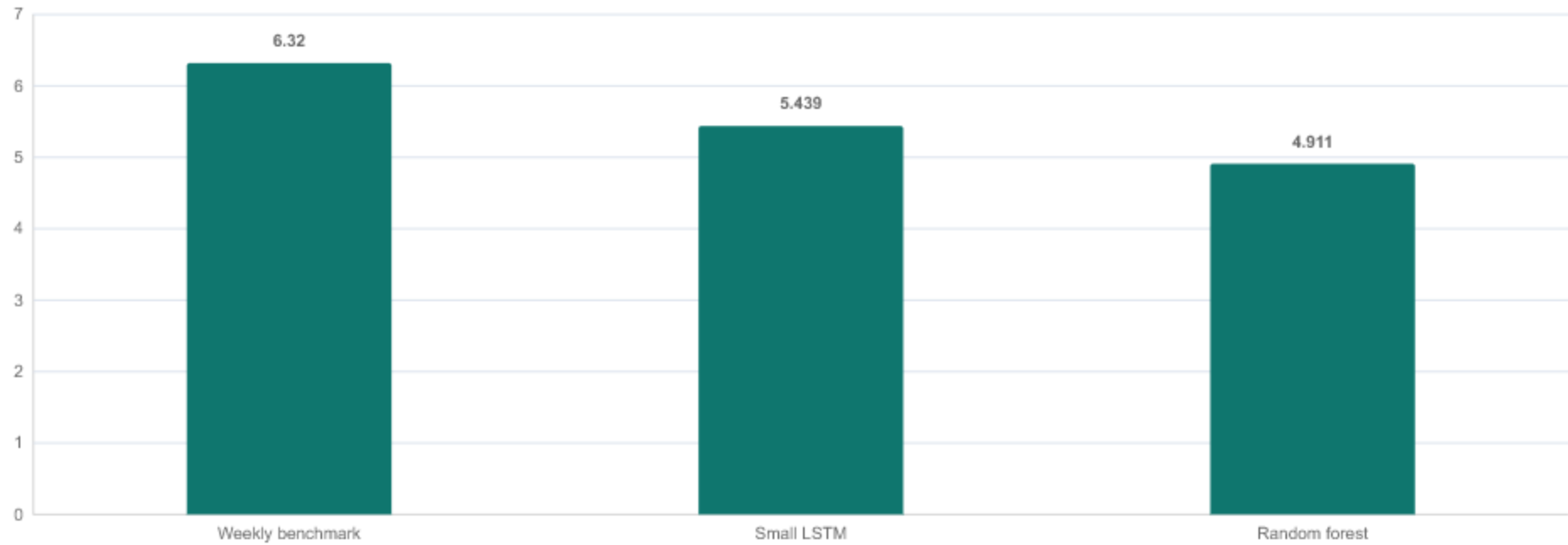
2013–2015: fit the first models

2016: validate and select by daily MAE

2017: final test using the frozen choice

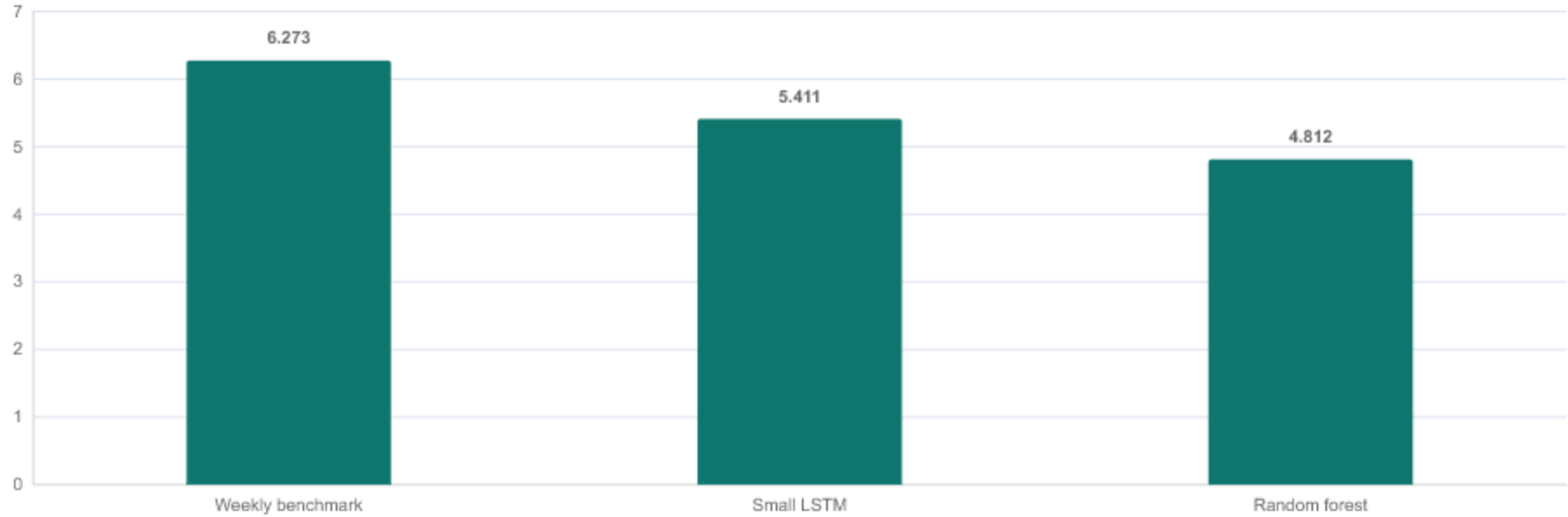
Compared: repeat last week, random forest and a small LSTM.

2016 validation chose random forest



Daily MAE in units. Lower is better.

2017 final test: daily and weekly trade-off



Random forest: 23.3% lower daily MAE than the benchmark.

LSTM had the lowest overall weekly-total MAE: 17.343 units.

Small LSTM comparison

28 past daily sales values

One 16-unit LSTM layer

Seven daily predictions

The LSTM improved weekly totals overall, but random forest had the lowest daily error.

Stock suggestion

Target stock = forecast during coverage + buffer

Suggested order = round up max(0, target − available − outstanding)

Example: 77.52 + 10 = 87.52 target units

With 50 available and 0 outstanding, suggest 38 units.

An early shortage warning is shown separately.

Ask StockSense

Five request types

Forecast, stock order, recent sales, model results and help

Text classification selects a request type

Answers use the same calculated values as the dashboard

Checks completed

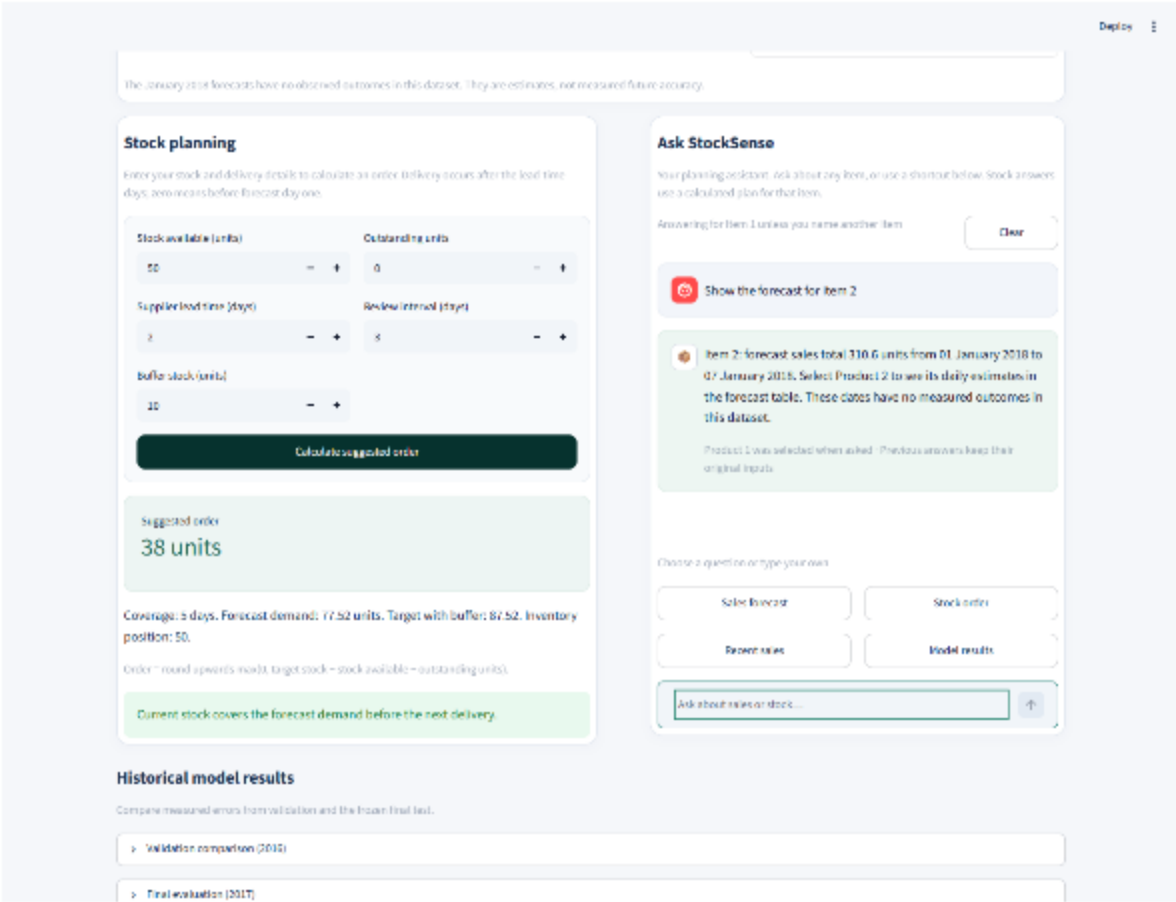
52 tests passed on the office PC

An earlier setup worked on a second PC.

Assistant check: 24/25 supported requests; 8/8 unrelated requests rejected.

Human usability results are still needed.

Live application demonstration



Select Product 1

Show seven dated estimates

Calculate the 38-unit example

Ask about another product

Findings and limits

The prototype gives a clear forecast and a transparent stock calculation.

Random forest performed best on daily error in this comparison.

Real savings and January 2018 forecast accuracy remain unmeasured.

Questions